

DOACROSS cost-model accuracy — 2 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.1947	0.1838	+6.0%	6.0%
Short chain	20,000	1	1	3.2964	2.4841	+32.7%	32.7%
Short, four chains	20,000	4	1	1.6180	1.6886	-4.2%	4.2%
Short, 64 chains	20,000	64	1	1.5864	1.4247	+11.4%	11.4%
Long chain	200,000	1	1	29.6727	23.2849	+27.4%	27.4%
Long, four chains	200,000	4	1	13.6303	12.8891	+5.8%	5.8%
Long, 64 chains	200,000	64	1	13.4537	13.3769	+0.6%	0.6%
Compute in dependency	32,768	1	1	10.1281	6.6943	+51.3%	51.3%
Compute before wait	32,768	4	1	4.4775	4.1467	+8.0%	8.0%
Compute after post	32,768	4	1	4.0350	3.5119	+14.9%	14.9%
Two dependent streams	32,768	4	2	4.2538	4.2261	+0.7%	0.7%
Uneven lanes + compute	8,193	3	1	1.4202	1.1636	+22.1%	22.1%
Large arrays, 64 chains	1,000,000	64	1	67.8487	65.3680	+3.8%	3.8%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before held-out timings; neither equation nor runtime changed.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 4 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.3359	0.2965	+13.3%	13.3%
Short chain	20,000	1	1	8.9352	3.1104	+187.3%	187.3%
Short, four chains	20,000	4	1	1.5333	0.9114	+68.2%	68.2%
Short, 64 chains	20,000	64	1	1.7389	0.9108	+90.9%	90.9%
Long chain	200,000	1	1	79.0435	27.1233	+191.4%	191.4%
Long, four chains	200,000	4	1	13.3709	7.7993	+71.4%	71.4%
Long, 64 chains	200,000	64	1	14.5403	8.5161	+70.7%	70.7%
Compute in dependency	32,768	1	1	24.1566	7.8460	+207.9%	207.9%
Compute before wait	32,768	4	1	3.6140	2.3542	+53.5%	53.5%
Compute after post	32,768	4	1	3.3262	2.1057	+58.0%	58.0%
Two dependent streams	32,768	4	2	3.9171	2.4822	+57.8%	57.8%
Uneven lanes + compute	8,193	3	1	1.3902	0.8397	+65.6%	65.6%
Large arrays, 64 chains	1,000,000	64	1	71.0033	43.2338	+64.2%	64.2%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

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DOACROSS cost-model accuracy — 6 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.5106	0.4595	+11.1%	11.1%
Short chain	20,000	1	1	14.0039	3.4164	+309.9%	309.9%
Short, four chains	20,000	4	1	2.4008	1.1786	+103.7%	103.7%
Short, 64 chains	20,000	64	1	2.2796	1.0564	+115.8%	115.8%
Long chain	200,000	1	1	144.8793	30.3985	+376.6%	376.6%
Long, four chains	200,000	4	1	23.1547	9.0221	+156.6%	156.6%
Long, 64 chains	200,000	64	1	19.1612	8.0691	+137.5%	137.5%
Compute in dependency	32,768	1	1	42.8658	8.2741	+418.1%	418.1%
Compute before wait	32,768	4	1	5.3918	2.2098	+144.0%	144.0%
Compute after post	32,768	4	1	5.1915	2.1201	+144.9%	144.9%
Two dependent streams	32,768	4	2	6.5116	2.6412	+146.5%	146.5%
Uneven lanes + compute †	8,193	3	1	n.a.	FAILED	n.a.	n.a.
Large arrays, 64 chains	1,000,000	64	1	80.7064	39.0056	+106.9%	106.9%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

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Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 8 workers

Learned synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.6334	0.5674	+11.6%	11.6%
Short chain	20,000	1	1	20.5781	3.9130	+425.9%	425.9%
Short, four chains	20,000	4	1	3.1268	1.2194	+156.4%	156.4%
Short, 64 chains	20,000	64	1	1.8157	0.9504	+91.0%	91.0%
Long chain	200,000	1	1	186.0166	34.5524	+438.4%	438.4%
Long, four chains	200,000	4	1	28.9306	7.7961	+271.1%	271.1%
Long, 64 chains	200,000	64	1	13.5234	7.4330	+81.9%	81.9%
Compute in dependency	32,768	1	1	58.3161	9.4984	+514.0%	514.0%
Compute before wait	32,768	4	1	4.2773	1.8537	+130.7%	130.7%
Compute after post	32,768	4	1	4.6484	2.0881	+122.6%	122.6%
Two dependent streams	32,768	4	2	7.8726	2.3331	+237.4%	237.4%
Uneven lanes + compute	8,193	3	1	2.8662	1.0945	+161.9%	161.9%
Large arrays, 64 chains	1,000,000	64	1	72.8401	37.7470	+93.0%	93.0%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before held-out timings; neither equation nor runtime changed.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 2 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.2479	0.1838	+34.9%	34.9%
Short chain	20,000	1	1	7.1796	2.4841	+189.0%	189.0%
Short, four chains	20,000	4	1	7.2085	1.6886	+326.9%	326.9%
Short, 64 chains	20,000	64	1	7.1559	1.4247	+402.3%	402.3%
Long chain	200,000	1	1	70.4578	23.2849	+202.6%	202.6%
Long, four chains	200,000	4	1	70.3846	12.8891	+446.1%	446.1%
Long, 64 chains	200,000	64	1	70.4304	13.3769	+426.5%	426.5%
Compute in dependency	32,768	1	1	14.5557	6.6943	+117.4%	117.4%
Compute before wait	32,768	4	1	13.6898	4.1467	+230.1%	230.1%
Compute after post	32,768	4	1	13.4731	3.5119	+283.6%	283.6%
Two dependent streams	32,768	4	2	22.6008	4.2261	+434.8%	434.8%
Uneven lanes + compute	8,193	3	1	3.5097	1.1636	+201.6%	201.6%
Large arrays, 64 chains	1,000,000	64	1	350.7599	65.3680	+436.6%	436.6%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

Times are medians of process medians. Predictions frozen before held-out timings; neither equation nor runtime changed.

Direct C probes use production model/dispatch/profiling APIs. † = failed output check or timeout; excluded from accuracy statistics.

DOACROSS cost-model accuracy — 4 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.3235	0.2965	+9.1%	9.1%
Short chain	20,000	1	1	7.2243	3.1104	+132.3%	132.3%
Short, four chains	20,000	4	1	7.2513	0.9114	+695.6%	695.6%
Short, 64 chains	20,000	64	1	7.2566	0.9108	+696.7%	696.7%
Long chain	200,000	1	1	70.5008	27.1233	+159.9%	159.9%
Long, four chains	200,000	4	1	70.3688	7.7993	+802.2%	802.2%
Long, 64 chains	200,000	64	1	70.4232	8.5161	+726.9%	726.9%
Compute in dependency	32,768	1	1	14.6769	7.8460	+87.1%	87.1%
Compute before wait	32,768	4	1	12.7590	2.3542	+442.0%	442.0%
Compute after post	32,768	4	1	12.6469	2.1057	+500.6%	500.6%
Two dependent streams	32,768	4	2	22.6025	2.4822	+810.6%	810.6%
Uneven lanes + compute	8,193	3	1	3.4070	0.8397	+305.8%	305.8%
Large arrays, 64 chains	1,000,000	64	1	352.3094	43.2338	+714.9%	714.9%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

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DOACROSS cost-model accuracy — 6 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.4285	0.4595	-6.8%	6.8%
Short chain	20,000	1	1	7.3414	3.4164	+114.9%	114.9%
Short, four chains	20,000	4	1	7.3433	1.1786	+523.0%	523.0%
Short, 64 chains	20,000	64	1	7.3288	1.0564	+593.8%	593.8%
Long chain	200,000	1	1	70.4481	30.3985	+131.7%	131.7%
Long, four chains	200,000	4	1	70.5600	9.0221	+682.1%	682.1%
Long, 64 chains	200,000	64	1	70.6030	8.0691	+775.0%	775.0%
Compute in dependency	32,768	1	1	14.8101	8.2741	+79.0%	79.0%
Compute before wait	32,768	4	1	12.4613	2.2098	+463.9%	463.9%
Compute after post	32,768	4	1	12.4104	2.1201	+485.4%	485.4%
Two dependent streams	32,768	4	2	22.6856	2.6412	+758.9%	758.9%
Uneven lanes + compute †	8,193	3	1	n.a.	FAILED	n.a.	n.a.
Large arrays, 64 chains	1,000,000	64	1	351.2917	39.0056	+800.6%	800.6%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

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DOACROSS cost-model accuracy — 8 workers

Seeded synchronization costs; full parallel loop, including launch and real wait/post dependencies.

Workload	Iterations N	Dependency distance	Sync streams	Predicted time (ms)	Measured time (ms)	Signed error (%)	Absolute error (%)
Tiny chain	257	1	1	0.4748	0.5674	-16.3%	16.3%
Short chain	20,000	1	1	7.4780	3.9130	+91.1%	91.1%
Short, four chains	20,000	4	1	7.4705	1.2194	+512.6%	512.6%
Short, 64 chains	20,000	64	1	7.4110	0.9504	+679.8%	679.8%
Long chain	200,000	1	1	70.6565	34.5524	+104.5%	104.5%
Long, four chains	200,000	4	1	70.5818	7.7961	+805.3%	805.3%
Long, 64 chains	200,000	64	1	70.5808	7.4330	+849.6%	849.6%
Compute in dependency	32,768	1	1	15.0855	9.4984	+58.8%	58.8%
Compute before wait	32,768	4	1	12.4381	1.8537	+571.0%	571.0%
Compute after post	32,768	4	1	12.4258	2.0881	+495.1%	495.1%
Two dependent streams	32,768	4	2	22.7725	2.3331	+876.1%	876.1%
Uneven lanes + compute	8,193	3	1	3.5351	1.0945	+223.0%	223.0%
Large arrays, 64 chains	1,000,000	64	1	351.9576	37.7470	+832.4%	832.4%

Signed error = 100 × (predicted / measured − 1). Negative = underprediction. Absolute error is its magnitude.

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